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From Pair Scores to Prerequisite Graphs on Skill-Taxonomy Descriptions

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We present a seven-step induction tool that converts pairwise large language model (LLM) prerequisite scores into an auditable directed acyclic graph (DAG). The evaluation uses ESCO-PrereqSkill under an unseen-concept split (49,608 test pairs; 2.0% positives). The only documents are the ESCO skill cards already present in a labels-only prompt (B1). On `gpt-4o-mini`, repeating those cards as retrieved spans (B3) raises ROC-AUC from 0.834 to 0.886 and PR-AUC from 0.085 to 0.099, yet lowers direction accuracy (0.761 to 0.667). The same format lowers AUC on Qwen 2.5 7B (0.803 to 0.654). A mention rule reaches PR-AUC 0.151, above every standalone LLM condition; a logistic stack of mention with B1 and the taxonomy prompt (B5) reaches 0.265. Temperature scaling has no bias term, so expected calibration error (ECE) barely changes (0.306 to 0.299). Reverse-conflict (C2) and cycle (C3) constraints on scaled B5 scores yield a DAG of 8,906 edges; E-test membership has pair F1 0.163 (332 of 1,003 gold), not the pair-level B5 F1 of 0.203. Candidate generation covers 48.3% of gold edges. We report when taxonomy-card formatting and directional constraints help, and when they do not.

Keywords: Prerequisite graph; large language models; skill taxonomy; evidence retrieval; calibration; directed acyclic graph.

1. Introduction

Prerequisite relations say which competence should be in place before another can be taught or assessed, and they therefore structure both learning paths and skill

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graphs. Recent pairwise predictors on educational corpora—including LECDPR [1] and ProPRL [2]—optimize a binary label $y_{ij} \in \{0, 1\}$ on a closed-set split and stop at a pair score. AutoPRE [3] and KnowLP [4] already use large language models (LLMs) for concept prerequisites or dual prerequisite–similarity graphs. The question that remains is narrower and more practical. Once those pair scores exist, when do taxonomy-card spans, temperature scaling, and directional constraints turn them into a graph that a practitioner can audit?

This paper introduces a seven-step induction tool for that conversion. We ablate each stage rather than claim that the full stack is reliable. On the public ESCO-PrereqSkill dump [5] we score candidates with `gpt-4o-mini` and, separately, with Qwen 2.5 7B Instruct. The headline protocol is the unseen-concept split of Zhu and Zamani: [6] any pair that touches a held-out skill is reserved for test. Because the test positive rate is only 2.0%, we report PR-AUC together with ROC-AUC. Lexical baselines and a mention-plus-LLM stack are evaluated on the same candidate pool. A restricted 240-pair human sample is reported in its own table.

The measurements support four contributions.

(i) We formalize an edge $e_{ij} = (s_i, s_j, \mathcal{E}_{ij}, u_{ij}, q_{ij}, t_{ij})$ with four type labels and a direction, and we specify a seven-step pipeline whose constraints C1–C5 are implemented in the reported run (C6 is specified but not applied).
(ii) On `gpt-4o-mini`, repeating the same ESCO cards under an Evidence heading improves ranking relative to a labels-only prompt (paired bootstrap ΔPR-AUC 0.014, interval 0.009–0.020). Adding the taxonomy path does not improve that ranking further.
(iii) The same format lowers direction accuracy on `gpt-4o-mini` and lowers ranking on Qwen 2.5 7B. A mention rule outperforms every standalone LLM condition on PR-AUC, yet a stack of mention with B1 and B5 recovers gold edges that mention misses and raises PR-AUC from 0.151 to 0.265.
(iv) Graph constraints applied to temperature-scaled B5 scores produce one directed acyclic graph. E-test pair F1 for membership in that graph is 0.163, which is not the pair-level B5 operating F1 of 0.203. The gain in forward–reverse margin comes from reverse-edge suppression on raw scores, not from cycle breaking.

We do not claim the first LLM prerequisite predictor, [3, 4] the first direction model, or that the graph stage outperforms evidence-only pair ranking under a shared validation threshold. A frozen zero-shot model is not trained on split A or split E, so the absence of an F1 drop from A to E is the expected outcome and is not treated as pretraining contamination. We do not generate learning paths [4, 7] and we do not use student traces. The rest of the paper is organized as follows. Section 2 reviews pairwise and taxonomy prerequisite work. Section 3 states the output graph. Section 4 presents the induction tool, the constraints C1–C5, and the ablation codes. Section 5 describes the ESCO dump, the splits, the models, and the

author-labeled sample. Section 6 reports pair ranking, graph recovery, calibration, and the open-family comparison. Section 7 discusses the conditional claim; Section 8 states the remaining limits of the evidence; Section 9 concludes.

2. Related Work

2.1. *Pairwise prediction on educational documents*

Early concept-prerequisite work measured Wikipedia and textbook signals (RefD), [8] recovered university course dependencies, [9] and learned concept graphs from educational click and catalog data. [10,11] Later extractors target massive open online courses (MOOCs), [12] online resources (PREREQ), [13] joint concept extraction (iPRL), [14] and course-description text. [15] Recent supervised and graph-neural models use learning-path supervision (LCPRE), [16] directed Weisfeiler–Leman message passing, [17] concept-pair GNNs, [18] and course-map convolution. [19] AutoPRE uses LLM agents to discover concept prerequisites. [3] KnowLP builds a dual graph of prerequisite and similarity edges for personalized learning paths. [4] LectureBank [20] is a slide-and-topic corpus for NLP education; the public dump we hold has topic titles only, so we omit it from the reported tables. LECDPR [1] retrieves rather than generates text, then applies a graph neural network over concept–document interactions. ProPRL [2] adds a behavioral view and a training loss L_{irr} that penalizes high scores in both directions. Retrieval-augmented generation retrieves then writes text; [21] we retrieve spans and never treat model-written text as evidence. Those earlier tables do not share a protocol with ours, so we do not treat pair F1 as a cross-paper ranking.

2.2. *Prerequisites on skill taxonomies*

KnowEdu constructs educational knowledge graphs from pedagogical data. [22] ESCO is the European skill classification; text-mining audits ask whether it tracks Industry 4.0. [23] Labor-market studies recover nested skill dependencies, [24] and known course prerequisites have been used to order modules toward jobs. [25] Those studies recover occupational or curricular structure; they do not produce a pedagogical edge with a retrieved span.

Le and Abel [5] released ESCO-PrereqSkill and compared zero-shot LLMs with expert links by generating prerequisite *skill names* and scoring them with semantic similarity and BERTScore. That is a generation-and-alignment task, not directed pair classification on a candidate pool, so we cite their published numbers only as a task description and not as a row in our tables. A later paper by Le, Abel and Laforge [26] tags learning resources with competencies, keeps an evidence span, and applies graph constraints on a competency graph. We score skill–skill pairs and do not tag documents. The two tools share the vocabulary of LLM scoring, evidence, and graph constraints, yet they solve different prediction problems.

2.3. *Unseen concepts and annotation*

Zhu and Zamani [6] showed that closed-set splits inflate apparent generalization. We therefore treat split E as the headline protocol. Textbook annotation protocols such as PREAP show that agreement depends on explicit criteria. [27] Our 240-pair sample uses a three-annotator queue rather than textbook PREAP, and it is too small to support a selective-review claim.

3. Problem

Let $V = \{s_1, \ldots, s_n\}$ be a skill set in which each node has a stable identifier, a label, a description, and, when present, a taxonomy path, and let a corpus of source documents be given. The output is a graph $G = (V, E)$ whose accepted edges are intended to be inspectable. Each edge carries retrieved evidence $\mathcal{E}_{ij}$ (a verbatim span together with a `source_id`), a confidence u_{ij}, a type $t_{ij} \in \{\text{direct}, \text{transitive}, \text{none}, \text{ambiguous}\}$, a direction $A \to B$, $B \to A$, or none, and an optional human-review state q_{ij}.

The LLM is used only as a bidirectional pair scorer $(p_{A\to B}, p_{B\to A})$. Each candidate pair is scored in one decoding at temperature 0. The two numbers are the model's own JSON fields for the stored orientation (source, target), not token log-probabilities, and they are discrete in steps of 0.1. Text written by the model is never stored as evidence. The graph keeps source $\to$ target when scaled $p_{A\to B} \geq \tau$ after C2; $p_{B\to A}$ is used to detect reverse conflict and is not inserted as a second edge unless (target, source) is itself a candidate.

Candidate generation does not enumerate $V \times V$, so graph-level recall cannot exceed gold-in-pool coverage. We report that ceiling explicitly.

4. The Induction Tool

4.1. *Seven steps*

Figure 1 summarizes the tool. Skills are first canonicalized with stable identifiers; the procedure does not invent nodes (S1). Candidates are then proposed from an inverted index and from label signals—shared bigrams, token containment, and token overlap of at least two—without reading gold edges (S2). For each ordered pair, S3 retrieves up to k=4 verbatim windows per direction from documents indexed by skill identifier or label. Windows are 280 characters long and are ranked by a heuristic that rewards order words (`before`, `requires`, `prerequisite`) and co-mention of the other skill.

S4 scores each candidate pair in one decoding at temperature 0 and requires JSON fields $p_{A\to B}$, $p_{B\to A}$, type, direction, `confidence_raw`, and `evidence_ids`. The system message and the three user-prompt variants appear in Appendix A. S5 fits temperature scaling [28] on the validation split. Platt scaling and isotonic regression were tried as pair-score maps; they are not used as the graph-stage map, because C2, C3, and C4 depend on an absolute probability scale (τ, μ=0.8, $|p_{A\to B} - p_{B\to A}| <$

0.05, and $c_{ij} > 0.8$). S6 applies the structural constraints below. S7 ranks pairs for human review under random, uncertainty ($1-|2p-1|$), and high-confidence policies.

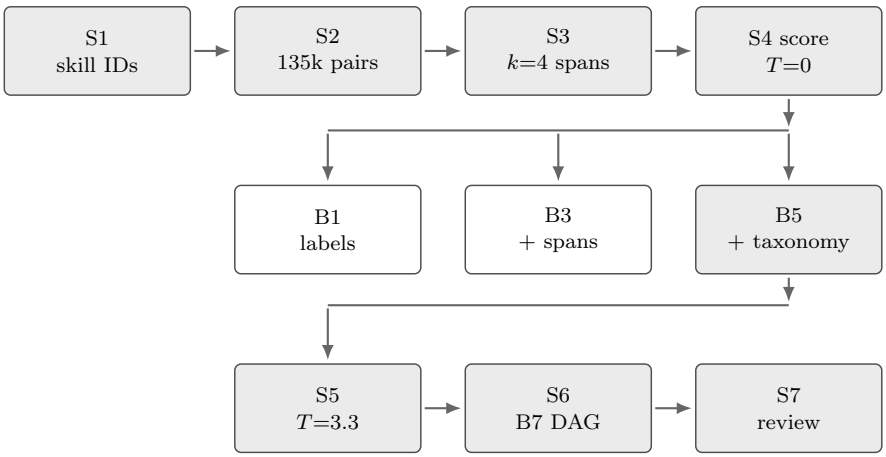


Fig. 1. Overview of the seven-step induction tool. Shaded boxes mark the reported path. B1, B3, and B5 are prompt variants at the scoring step (S4). Only B5 is temperature-scaled (S5) and constrained (S6) into the reported DAG; B6 and B7 transform those B5 scores without a further model call. C4 flags edges; C5 is never triggered on this dump.

4.2. *Constraints C1–C6 and* L_{irr}

Let τ be the operating threshold and let μ=0.8. The reported run implements C1–C5; C6 is specified so that the type field is well defined, but it is not applied to the pair tables.

- C1 (self-loop). Drop any pair with $s_i = s_j$.
- C2 (reverse conflict). If $p_{A\to B} > \tau$ and $p_{B\to A} > \tau$, flag the pair. If $|p_{A\to B} - p_{B\to A}| < 0.05$, mark a direction conflict; otherwise suppress the weaker direction to $\min(p_{\text{weak}}, \max(0, \mu - p_{\text{strong}}))$.
- C3 (cycles). Build a directed graph of pairs with $p_{A\to B} \geq \tau$ after C2. While a strongly connected component of size greater than one remains, sort its internal edges by uncertainty $1 - c_{ij}$, where c_{ij} is the temperature-scaled `confidence_raw` of that pair, and delete a batch of $k = \max(1, \min(250, \lceil m/20 \rceil))$ edges (m is the number of internal edges). Lowest c_{ij} is deleted first. The procedure is batched lowest-confidence deletion, not a minimum feedback arc set. At most 80 batched rounds are run; leftover cycles, if any, are removed one edge at a time until the graph is acyclic.
- C4 (taxonomy). Flag a high-confidence pair ($c_{ij} > 0.8$, same scaled `confidence_raw`) whose top-level taxonomy paths differ. The flag does not by itself delete the edge. On the reported full-pool run, C4 flags 178 pairs.

- C5 (no evidence). Flag a pair that has no retrieved span in either direction. C5 is never triggered on this dump (C1 self-loops 0; no-evidence 0), because every scored pair has an ontology-card window.
- C6 (type). Intended to separate direct from transitive edges after the graph is built. It is not applied in the pair tables. A post-hoc transitive reduction of the gold graph finds only 38 shortcut edges (0.7% of 5,797 unique gold pairs). On the accepted B7 DAG, 1,534 of 8,906 edges (17.2%) have an alternate path after the edge is removed; only 169 of those 1,534 are gold-direct. C6 would therefore relabel a non-trivial slice of the predicted graph, but that slice is mostly not gold. A second 240-pair human sheet oversampled alternate-path candidates from the earlier val+test DAG of 6,178 edges (Section 5.5); annotators assigned `transitive` only once, and never on a gold shortcut or a B7 alternate-path edge. C6 remains a graph heuristic, not a human type.

The quantity L_{irr} is not a training loss in this work: there is no gradient step. Following the irreversibility penalty of ProPRL, [2] we compute the post-hoc mean $L_{\text{irr}} = \text{mean}\max(0, p_{A\to B} + p_{B\to A} - \mu)$ on gold-positive pairs in the scored set (μ=0.8). C2 is the corresponding inference rule.

Reverse-conflict rate (RCR) is

$$\text{RCR} = \frac{|\{i : \text{C2 flag}\}|}{|\{i : p_{A\to B} \geq \tau\}|}.$$

The C2 flag is not cleared after C3, so RCR after the graph is acyclic does not mean that the accepted DAG still contains two-way edges. The accepted graph keeps at most one direction per pair. On the reported full-pool run, C2 flags 10,699 pairs (RCR 10,699/31,670 $\approx$ 0.338 at scaled τ=0.55). C3 then deletes 22,764 edges; after the batched pass, 1,303 leftover cyclic edges were removed over 140 sequential rounds. We do not interpret post-S6 RCR as residual reverse edges.

4.3. *Ablations*

Table 1 defines the conditions that we compare. Codes B2 and B4 were reserved for a lexical-only scorer and a supervised pair classifier in an earlier design; they were not run as LLM conditions. The mention and TF-IDF baselines in Section 6 play the role that B2 would have played. B6 and B7 are post-hoc transforms of the B5 scores; they do not call the LLM again. S4-lite applies only C2 to B5 scores.

4.4. *Metrics*

At the pair level we report precision, recall, and F1 at a validation-chosen threshold τ, together with F1 at the fixed cut 0.5 as a diagnostic, ROC-AUC, and PR-AUC (average precision). With a 2.0% positive rate, PR-AUC is the ranking metric we emphasize. Direction, reported only on gold $A \to B$ pairs, is measured by direction accuracy (DA) and by the mean forward–reverse margin (FR-margin, mean $p_{A\to B} -$

Table 1. Ablation conditions. B6 and B7 transform B5 scores without a further model call. B2 and B4 were not run as LLM conditions.

Code	Evidence	Taxonomy prompt	Calibration	Graph
B1	no	no	no	no
B3	yes	no	no	no
B5	yes	yes	no	no
B6	yes	yes	S5	no
B7	yes	yes	S5	S6
S4-lite	yes	yes	no	C2 only

$p_{B\to A}$). At the graph level we record whether the accepted graph is a directed acyclic graph; the numbers of C2 flags, C3 deletions, C4 taxonomy flags, and C5 no-evidence flags; and directed precision and recall of accepted edges against unique gold edges (full set, in-pool, and scored Etv subset). Accepted edges are also compared with the transitive closure of the gold graph. Calibration uses expected calibration error (ECE, ten equal-width bins) and Brier score on the JSON probabilities. Selective review reports binary accuracy against human gold on the leading b% of the 240-pair sample. All LLM scores come from a single decoding at temperature 0. Pair ROC-AUC, PR-AUC, and F1@τ on E test are reported with 95% bootstrap intervals from 1,000 resamples of the 49,608 pairs (seed 42).

5. Experimental Setup

5.1. *Data and documents*

The primary corpus is the public ESCO-PrereqSkill dump. [5] After conversion we have 5,628 skill nodes, 5,817 gold-edge records (self-loops dropped; 5,797 unique directed pairs after collapsing 20 duplicate rows), and 5,627 documents. The source paper describes 3,196 skills. The difference is mechanical rather than a claim of a new dataset: the public CSV has 3,759 target rows, and resolving prerequisite names against ESCO preferred and alternate labels adds endpoints that are not themselves target rows, producing 5,628 nodes.

Each document is the skill's own preferred label together with its ESCO description (`source_type=ontology`). One unresolved name-only node has no description and no document, which is why there are 5,627 documents for 5,628 skills. Retrieved evidence is therefore a window over taxonomy text, not an independent textbook or lecture. B3 and B5 both see that text: B3 as retrieved spans, B5 as those spans plus the taxonomy path in the prompt, and B1 as the raw label and description already present in the skill record. The ablation therefore concerns *how* the description is shown, not a second corpus.

Candidate generation yields 135,726 pairs, and gold-in-pool coverage is $2{,}811/5{,}817 = 0.483$ gold records (unique in-pool gold in Table 5 is 2,800 of 5,797). Any graph built only from this pool has recall at most 0.483 against the full gold

edge set, even before thresholding.

5.2. *Splits*

Split A is a random 8:1:1 split of candidate pair identifiers (108,580 / 13,572 / 13,574) and is used only as a pair-score reference. Split E is the headline protocol. [6] With seed 42 we hold out 20% of the skill identifiers that appear in the candidate pool (1,125 skills). Every pair that touches a held-out skill is reserved for test (49,608 pairs, about 36% of the pool, 1,003 gold positives). The remaining pairs are split 90:10 into train and validation (77,507 / 8,611). We do not train a scorer on the 135,726-pair pool, and we do not train the graph networks of LECDPR or ProPRL. A frozen zero-shot scorer never sees split-A training pairs, so split membership cannot change its parameters.

5.3. *Models and decoding*

The headline family is `gpt-4o-mini`, scored under B1, B3, and B5 on E validation and E test; B5 is also scored on the 77,507 E-train pairs that enter the reported graph. The open family is Qwen 2.5 7B Instruct, served through an OpenAI-compatible API and scored under the same three prompts on E validation and E test. Decoding temperature is 0, and the probabilities are the JSON fields $p_{A\to B}$ and $p_{B\to A}$. The two families are reported in separate test tables. Token use and wall-clock time appear in Appendix B.

5.4. *Lexical baselines*

On the same 49,608 E-test pairs we score three non-LLM rules: Jaccard overlap of label-and-description tokens (the S2 similarity field); TF-IDF cosine of the same text (unigrams and bigrams); and a directed mention rule that returns 1 if the source preferred label occurs in the target description and 0 otherwise. The mention rule is a diagnostic of surface overlap in the dump, not a pedagogical oracle. Many gold links in ESCO-PrereqSkill pair a knowledge label with an action that names that label—for example, *musical notation* before *transcribe ideas into musical notation*. A supervised pair classifier is fit only on E-train pairs (77,507 rows, 1,605 positives). The document is the concatenation of the two skill texts with a separator; the vectorizer is TF-IDF unigrams and bigrams (80,000 features) fit on train only. We report an unweighted logistic regression, a class-balanced logistic regression, and a linear SVM. The validation split chooses τ; the test split is unused until the final row.

Classic RefD [8] needs Wikipedia in- and out-links. An official Wikidata P4644 map yields 523 of our 5,627 http skills with an English Wikipedia sitelink, but only 760 E-test pairs and five gold positives have both endpoints mapped, so a Wikipedia RefD ranking is not available on this dump. Two structural substitutes are scored on the same E-test pairs. A taxonomy-prefix score is nearly vacuous:

5,627 of 5,628 skills have a path of length exactly two, and gold edges often cross those two top-level branches (AUC 0.163, PR-AUC 0.020). A mention-graph analog of RefD—the related set is the skills whose description contains the other label, after labels shorter than six characters are dropped—reaches PR-AUC 0.141 and F1 0.311 at τ=0.05. That construction is the mention rule in other clothing, not a second corpus, and it does not outperform the one-line mention check.

5.5. *Human sample*

Two annotators and one adjudicator—the three authors—labeled 240 pairs with $\{0, 1, 2, 3\}$ (none, direct, transitive, ambiguous) plus a direction, without seeing gold or model scores. Cohen's κ on the 80-pair overlap is therefore in-lab author agreement, not an independent-rater estimate. The sheet showed labels, descriptions, taxonomy paths, and retrieved spans. Sampling (seed 42) mixed 80 overlapping pairs for agreement, 80 unique pairs per annotator, and a 15-pair calibration block drawn from a 200-pair stratified inspect set. The overlapping block itself mixed uncertainty, random, gold-positive, and C2-flagged pairs. On those 80 overlapping pairs the raw agreement is 0.875 and Cohen's κ is 0.79. The adjudicated gold has 151 none, 58 direct, 31 ambiguous, and 0 transitive labels. For binary metrics we treat direct as positive and both none and ambiguous as negative (n_+=58).

A second 240-pair sheet (seed 42, no overlap with the first) oversampled the 38 gold shortcuts and the B7 edges that have an alternate path, and required an intermediate skill name whenever the label was `transitive`. Raw agreement on the new 80-pair overlap is 0.812 and κ=0.72. The adjudicated labels are 81 none, 77 direct, 81 ambiguous, and 1 transitive. None of the 31 gold shortcuts in that sheet, and none of the 160 C6-oversampled pairs, received a human `transitive` label. The single transitive sits on a gold-direct pair without a B7 alternate path. The type field t_{ij} is therefore still not scored as a four-way classifier: the human prior on `transitive` is near zero even under oversampling, so C6 cannot be checked as a type detector. Accuracy at budget b is binary accuracy of the pair score at threshold 0.5 against this human gold, computed only on the leading $\lfloor 0.01\, b \cdot 240 \rfloor$ pairs under a ranking policy. At b=10 that slice has 24 pairs; a gap of 0.012 is smaller than one pair and is not a statistical claim.

6. Results

Pair scores in this section come from a single temperature-zero decoding of each named condition on the held-out split. A 50-pair connectivity check and a 200-pair inspect set used during development are excluded from the test tables.

6.1. *Pair ranking on ESCO split E*

Table 2 is the headline comparison. On `gpt-4o-mini`, presenting the skill cards as retrieved spans raises ROC-AUC from 0.834 to 0.886 and PR-AUC from 0.085 to

0.099. Inserting the taxonomy path on top of those spans does not improve either ranking metric beyond B3. F1 at the fixed cut 0.5 falls from B1 to B3 because the spans make the model attach edges too densely (recall 0.98, precision 0.041). That cut is the wrong operating point under a 2% positive rate, and it is not used as the headline metric.

Validation-chosen F1 is available for every LLM condition. For B1, temperature scaling on validation keeps τ=0.5 and test F1 remains 0.163. For B3 and B5, the same step followed by τ=0.55 yields test F1 0.190 and 0.203. The reported graph condition, B7, starts from those B5 scores, rescales both $p_{A\to B}$ and $p_{B\to A}$ with T=3.3, and then applies the reverse-conflict and cycle constraints. Pair-level B5 F1@τ (0.203, recall 0.593) thresholds temperature-scaled `confidence_raw` on the 49,608 E-test rows. Table 5 uses a different predicate: membership in the directed edge set with scaled $p_{A\to B} \geq 0.55$ after C2. On that edge set, test recall is 0.886 (889 of 1,003). After C3, 332 of those gold test edges remain, so E-test pair F1 for DAG membership is 0.163 (precision 0.108, recall 0.331, 3,079 test pairs inside the DAG). The two F1 figures are therefore not a before-and-after of one cut. A further diagnostic, S4-lite, applies only C2 to raw B5 scores (F1 0.126) and is not used as an operating point.

The mention rule reaches PR-AUC 0.151 and F1@0.5=0.339, above every standalone LLM condition on this dump. TF-IDF cosine is a strong ROC ranker (AUC 0.808) but a weak precision–recall ranker (PR-AUC 0.070). Jaccard overlap is near chance. A linear SVM on concatenated skill text, fit on E train, reaches PR-AUC 0.073 and F1@τ=0.30 of 0.111. Unweighted logistic regression never predicts a positive on validation (F1 0); class-weighted logistic regression reaches only PR-AUC 0.051. Neither supervised model outperforms the mention rule, or B3 and B5, on PR-AUC. The mention-graph RefD analog (PR-AUC 0.141) stays just below mention, and the taxonomy-prefix score is worse than chance.

A binary union—mention or B5 at τ=0.55—is the wrong hybrid: it raises recall to 0.616 but drops precision to 0.126 and PR-AUC to 0.085. The useful combination is a logistic stack fit on validation (mention, raw B1 score, raw B5 score). That stack reaches PR-AUC 0.265 and ROC-AUC 0.896; at the validation F1 cut τ=0.15 it has precision 0.458, recall 0.256, and F1 0.328. The stack therefore wins on ranking, not at the operating point: mention F1@0.5 remains higher (0.339). The stack weights and τ were chosen on the same 8,611-pair validation split; a cross-validated stack is not reported. B5 at τ=0.55 recovers 373 of the 758 gold test edges that mention misses, while mention alone keeps 24 gold edges that B5 drops. At the mention recall of 0.244, mention precision is 0.556 and B5 precision is only 0.131; the stack sits at 0.519. Precision at 100 and at 1,003 follows the same order (stack 0.65 / 0.32, mention 0.53 / 0.26, B5 0.07 / 0.13). The ranking gain of B3 over B1 is therefore real on `gpt-4o-mini` and still smaller, in standalone PR-AUC, than mention. The LLM scores become useful for ranking only after they are combined with that rule.

A shuffled-card placebo isolates format from content. On a gold-complete E-test sample (all 1,003 positives and 4,012 random negatives; prevalence 20%), B3 with

the true skill cards has ROC-AUC 0.887 and PR-AUC 0.562. The same prompt with evidence taken from another pair drops to 0.685 / 0.394, below labels-only B1 on that sample (0.836 / 0.506). Repeating the matched ESCO card under an Evidence heading improves ranking; a mismatched card lowers both metrics. Those three rows are reported separately from Table 2 because the sample prevalence is not 2%.

Table 3 gives 95% bootstrap intervals. Independent intervals for B3 and B1 PR-AUC nearly touch (0.077–0.093 versus 0.092–0.108). A paired bootstrap of the difference on the same 49,608 pairs gives ΔROC-AUC 0.052 (0.043–0.062) and ΔPR-AUC 0.014 (0.009–0.020); both differences are positive in all 1,000 resamples. The mention interval (0.130–0.175) stays above every standalone LLM interval. B5 F1@τ (0.190–0.217) overlaps B3 (0.179–0.202), so the 0.203 versus 0.190 gap is not a stable ranking win.

Table 2. `gpt-4o-mini`, lexical rules, and a linear SVM on ESCO split E test (49,608 pairs, 1,003 positives). F1 at 0.5 is a diagnostic. F1@τ uses the validation-chosen cut after temperature scaling, except S4-lite (raw B5 at 0.55), the mention rule (binary, τ=0.5), the stack (validation F1 sweep), and the SVM (validation F1 sweep). Lexical and SVM rows have no temperature step. B7 F1@τ is E-test pair membership in the reported DAG (3,079 of 49,608 pairs), not graph-level F1 on all 8,906 edges. B7 DA is the score-level figure on all gold test pairs (0.715, equal to B5); on the 332 gold test edges inside the DAG it is 0.569.

Condition	F1@0.5	F1@τ	τ	AUC	PR-AUC	DA
Mention (label in desc.)	0.339	0.339	0.50	0.620	0.151	—
TF-IDF cosine	0.044	—	—	0.808	0.070	—
Jaccard tokens	0.005	—	—	0.564	0.027	—
B1 labels only	0.163	0.163	0.50	0.834	0.085	**0.761**
B3 + spans	0.079	0.190	0.55	0.886	0.099	0.667
B5 + taxonomy	0.071	0.203	0.55	0.880	0.101	0.715
Stack mention+B1+B5	—	**0.328**	0.15	**0.896**	**0.265**	—
B7 + DAG	—	0.163	0.55	—	—	0.715
S4-lite B5+C2 (raw)	—	0.126	0.55	—	—	—
Linear SVM (TF-IDF)	0.014	0.111	0.30	0.697	0.073	—

Table 3. Bootstrap 95% intervals on E test (1,000 resamples, seed 42). Point estimates match Table 2. The B5 AUC interval is centered at 0.879 rather than the 0.880 in that table because eight initially failed requests were scored as zero in the bootstrap file.

Condition	AUC	PR-AUC	F1@τ
Mention	[0.608, 0.634]	[0.130, 0.175]	[0.309, 0.372]
B1	[0.820, 0.846]	[0.077, 0.093]	[0.152, 0.175]
B3	[0.878, 0.893]	[0.092, 0.108]	[0.179, 0.202]
B5	[0.871, 0.888]	[0.093, 0.111]	[0.190, 0.217]

B1 F1 is 0.163 on E and 0.156 on A. Because B1 is a frozen zero-shot scorer, the two splits cannot change its weights, and the absence of a drop is the expected outcome rather than a membership test. We do not read it as pretraining contamination. A check that the ranking metrics can fail is label permutation of the E-test scores (20 shuffles, seed 42). ROC-AUC falls to 0.50 (interval 0.49–0.51) and PR-AUC falls to the 2.0% base rate for B1, B3, B5, and the mention rule. That diagnostic shows that the metrics respond to destroyed labels; it is not a membership test. A post-cutoff concept slice is not available on this dump. ESCO v1.2 (May 2024) is after the October 2023 knowledge cutoff of `gpt-4o-mini`, but a URI set-difference of our 5,627 http skills against the public ESCO v1.1.1 list (September 2022) yields one new identifier, ten E-test pairs, and zero gold positives. Last-modified dates in the current ESCO CSV cannot replace that test: 10,770 of 13,960 skills carry a 2024 stamp after the v1.2 republish.

6.2. *Worked pairs*

Table 4 shows four inspect pairs from the split-E scores that are not in the 240-pair human sheet. Probabilities are the JSON fields $p_{A\to B}/p_{B\to A}$, not token log-probabilities. Spans are the first retrieved $A \to B$ window (character limit 280).

The mention pattern is already easy without retrieval: *soldering techniques* before *apply soldering techniques* is 0.8/0.8 under B1 and only slightly higher under B3 and B5, with both directions high. Evidence can also rescue a gold pair that B1 misses: *types of concrete pumps* before *install concrete pumps* moves from 0.0/0.0 to 0.7/0.5. The same spans inflate a same-verb distractor (*assemble mechatronic units* / *assemble hardware components*, gold none) from 0.3/0.2 to 0.7/0.5, and they raise both directions on a gold-reverse pair (*design user interface* before *use methodologies for user-centered design*). The four rows are illustrations, not a sample statistic.

6.3. *Error patterns*

A stratified read of 200 inspect pairs, together with an automatic audit of the 3,006 gold edges that S2 never proposed, shows three recurring patterns.

Containment and mention. Many gold $A \to B$ edges pair a knowledge label with an action that names that label (*soldering techniques* / *apply soldering techniques*; *musical notation* / *transcribe ideas into musical notation*). The mention baseline in Table 2 is strong because the dump is written that way, not because surface overlap is a theory of pedagogy. The current candidate generator already uses token containment: an automatic miss audit finds 0 remaining gold edges whose one label is a token-subset of the other, 22 missed edges that share a bigram, and 1,548 missed edges with at least one overlapping token. Adding those 1,548 single-token pairs to the current pool would raise gold coverage from 0.483 to 0.749. The leftover misses are mostly same-domain, low-overlap pairs (*computer programming* / *use concurrent programming*; *toxicology* / *perform toxicological studies*).

Table 4. Four inspect pairs scored on split E. Values are $p_{A\to B}/p_{B\to A}$. None of the four pairs is in the 240-pair human sheet.

P1. *soldering techniques → apply soldering techniques.* Gold: direct (label containment). Span: "apply soldering techniques. Apply and work with a variety of techniques in the process of soldering, such as soft soldering, silver soldering, induction soldering. . . " B1 0.8/0.8; B3 0.9/0.8; B5 0.9/0.8.
P2. *types of concrete pumps → install concrete pumps.* Gold: direct. Span: "types of concrete pumps. The types of machines used to pump liquid concrete such as the boom concrete pumps used for large construction projects or line pumps. . . " B1 0.0/0.0; B3 0.7/0.5; B5 0.7/0.5.
P3. *assemble mechatronic units → assemble hardware components.* Gold: none (same verb, different object). Span: "assemble mechatronic units. Assemble mechatronic units using mechanical, pneumatic, hydraulic, electrical, electronic, and information technology systems and componen[ts]. . . " B1 0.3/0.2; B3 0.7/0.5; B5 0.6/0.4.
P4. *design user interface → use methodologies for user-centered design.* Gold: none (reverse of an ontology edge). Span: "design user interface. Create software or device components which enable interaction between humans and systems or machines, using appropriate techniques. . . " B1 0.6/0.4; B3 0.8/0.7; B5 0.7/0.7.

Same verb, different object. Candidate negatives often share a verb and differ in the object (*assemble mechatronic units / assemble hardware components*; *process returned timber products / process dairy farm products*). B1 is usually conservative on those pairs. B3 and B5 raise $p_{A\to B}$ when the retrieved window repeats the shared verb, which is the density problem behind the drop in F1@0.5.

Reverse action → knowledge. Ontology gold, when it exists, is typically knowledge before action. The reverse pair is labeled none. Models still assign mass in both directions once a shared span is shown (P4 in Table 4). C2 is meant to suppress the weaker side of that conflict; it does not invent the missing gold direction.

6.4. *Direction and structure*

Evidence lowers direction accuracy on gold $A \to B$ pairs: B1 0.761, B5 0.715, B3 0.667 (FR-margin 0.178, 0.123, 0.126). On the reported scaled path, B5 produces 10,699 C2 flags. C3 then deletes 22,764 edges and leaves a DAG of 8,906 accepted edges. C4 flags 178 high-confidence pairs whose top-level taxonomy paths differ and does not delete them; C5 flags none. Score-level direction accuracy of B7 equals that of B5 (0.715), because temperature scaling is monotone. Restricted to the 332 gold test edges that remain in the DAG, direction accuracy falls to 0.569. The raw forward–reverse margin stays 0.123; the 0.395 figure belongs to S4-lite (C2 on raw $p_{A\to B}$ and $p_{B\to A}$) and is not a property of the scaled DAG. The graph stage therefore yields acyclicity; it does not improve pair F1.

6.5. *Graph-level recovery*

The accepted DAG is induced from the 135,726 scored E train, validation, and test pairs. Of 1,605 in-pool train gold edges, 501 appear in B7. Table 5 therefore reports three gold denominators on the 5,797 unique directed gold pairs, all computed from the same temperature-scaled B5 scores.

B7 keeps 8,906 edges with precision 0.100 against unique gold. Recall is 0.154 on the full gold set, 0.318 on gold-in-pool, 0.326 on the Etv gold subset (389 of 1,195), and 0.331 on E test (332 of 1,003). Scoring the train pairs raises full-gold and pool recall relative to the validation-and-test DAG (0.096 / 0.199), but C3 on the denser graph lowers Etv recall from 0.465 to 0.326 and test recall from 0.423 (424 true positives) to 0.331. B5 at scaled $p_{A\to B} \geq 0.55$ after C2, before cycle breaking, is denser (31,670 directed edges) and recovers more gold (recall 0.423 full / 0.875 in-pool / 0.886 test) at lower precision (0.077). The 0.483 S2 ceiling still bounds any graph built only from this pool; B7 reaches 0.318 of gold-in-pool because C3 deletes true positives. If S2 also kept label pairs that share a single token, coverage would rise to 0.749 (4,359 of 5,817 gold edges) at about 562k directed pairs—a larger pool, not a free gain.

Of the 8,016 B7 edges that miss gold-direct, only three sit on the gold transitive closure: they are not correct merely by being transitive. The closure of B7 does reach 1,219 gold edges (0.210 of unique gold), so some missed gold links are implied by predicted paths. The gold graph itself is already nearly reduced: only 38 of 5,797 unique gold edges are transitive shortcuts (0.7%). That is why C6, had it been applied, would rarely relabel a gold edge on this dump.

Table 5. Directed graph recovery from one full-pool run (135,726 scored pairs; 5,797 unique gold). B5@τ is scaled $p_{A\to B} \geq 0.55$ after C2, before C3. B7 is the accepted DAG on the same scores. R test uses the 1,003 gold E-test edges (889 and 332 true positives). This is not the pair-level B5 recall 0.593 in Table 2, which thresholds scaled `confidence_raw`.

Graph	$\|E\|$	Prec.	R full	R pool	R Etv	R test
B5@τ (directed)	31,670	0.077	0.423	0.875	0.884	0.886
B7 accepted DAG	8,906	0.100	0.154	0.318	0.326	0.331

6.6. *Calibration*

Table 6 reports ECE and Brier score on E test. B1 is already the best-calibrated JSON scorer (ECE 0.086). Temperature scaling makes B1 worse (ECE 0.109). On B3 and B5 the same step moves ECE from 0.282 to 0.273 and from 0.306 to 0.299. That flatness is expected: temperature scaling divides the logit by T and has no intercept, so it cannot undo a systematic over-positive mean when the base rate is 2%. On B5, Murphy resolution falls from 0.0011 to 0.0005 while reliability stays

large (0.160 to 0.141). Platt scaling and three-parameter beta calibration add a bias term and drive ECE to 0.0015 and 0.0014 by mapping almost every score to the 2% base rate (resolution 0.0003 and 0). Because those maps are monotone, a threshold rechosen on validation would leave pair F1 and AUC unchanged. We therefore do not reject Platt as a pair-ranking calibrator. We still keep temperature scaling as the implemented S5 step for the graph: C2, C3, and C4 use absolute cutoffs (τ=0.55, μ=0.8, $|p_{A\to B} - p_{B\to A}| < 0.05$, $c_{ij} > 0.8$), and a map that collapses to the base rate makes those constants meaningless. The JSON scores themselves are discrete in steps of 0.1, so many pairs share a rank; that granularity affects ECE, AUC, and the order in which C3 deletes edges. The calibration stage does not support the phrase "uncertainty-calibrated" as an empirical claim.

Table 6. Calibration on ESCO split E test. Temperature T is fit on validation. Probabilities are JSON fields, not log-probabilities.

Condition	T	ECE raw	ECE cal.	Brier raw / cal.
B1	3.2	**0.086**	0.109	0.055 / 0.064
B3	3.9	0.282	0.273	0.174 / 0.146
B5	3.3	0.306	0.299	0.178 / 0.160

6.7. *Open family*

Table 7 compares the two families on the same E validation split (8,611 pairs). On Qwen, evidence lowers AUC from 0.814 to 0.652; B5 recovers part of that drop (0.710) but does not outperform B1. Direction accuracy moves in the opposite direction of ranking (DA 0.839 to 0.943).

Table 8 is the matching Qwen E-test run (49,608 pairs). The pattern repeats: AUC falls from 0.803 (B1) to 0.654 (B3); B5 recovers to 0.696 and still stays below B1. PR-AUC follows the same order (0.062, 0.032, 0.036), while DA rises (0.809 to 0.908 to 0.947). Whether retrieved cards help ranking is therefore family-dependent; the `gpt-4o-mini` test numbers remain in Table 2.

6.8. *Human sample*

On 240 human labels, uncertainty sampling reaches accuracy 0.808 versus random 0.796 at a ten percent budget (24 pairs). High-confidence sampling reaches 0.813. The gaps are smaller than one pair and are not interpreted as a gain for selective review.

Table 9 compares the same scores with those 240 labels. The slice is denser in positives (58 of 240) than the ontology. B3, B5, and Qwen-B3 have recall 1.0 at threshold 0.5. Human PR-AUC favors B5 (0.748) over B1 (0.634), which matches the ranking pattern of `gpt-4o-mini` and does not replace Table 2.

Table 7. ESCO split E validation (8,611 pairs); F1 at 0.5.

	Qwen F1 / AUC / DA	mini F1 / AUC / DA
B1	0.067 / **0.814** / 0.839	**0.166** / 0.832 / 0.750
B3	0.045 / 0.652 / 0.901	0.083 / **0.874** / 0.641
B5	0.044 / 0.710 / **0.943**	0.076 / 0.864 / 0.708

Table 8. Qwen 2.5 7B on ESCO split E test (49,608 pairs); F1 at 0.5.

Condition	F1@0.5	AUC	PR-AUC	DA
B1 labels only	**0.061**	**0.803**	**0.062**	0.809
B3 + spans	0.041	0.654	0.032	0.908
B5 + taxonomy	0.040	0.696	0.036	**0.947**

Table 9. Same 240 author-labeled pairs (positive = direct; none and ambiguous = negative). Not a substitute for Table 2.

Condition	F1@0.5	F1@0.55	AUC	PR-AUC
mini B1	**0.714**	**0.768**	0.872	0.634
mini B3	0.435	0.641	0.899	0.612
mini B5	0.494	0.699	**0.949**	**0.748**
Qwen B3	0.395	0.459	0.720	0.550

7. Discussion

The measurements support a conditional claim about this dump. Repeating an ESCO skill card under an Evidence heading helps `gpt-4o-mini` rank pairs and lowers its direction accuracy. The same format lowers Qwen ranking on both validation and the 49,608-pair test set. Every one of the 49,608 B3 test spans is the card of skill A or B, so there is no independent corpus. A shuffled-card control still matters: on a gold-complete subsample the matched card outperforms B1, and a card from another pair falls below B1. The gain is therefore not a generic Evidence heading. That is why B3 and B5 can look alike, and why a mention rule is strong.

Standalone LLM PR-AUC stays below mention (0.151; bootstrap 0.130–0.175 versus at most 0.093–0.111 for B5). A linear SVM on concatenated TF-IDF also stays below mention (PR-AUC 0.073). The LLM scores still add edges that mention never sees: a validation-fit stack of mention with B1 and B5 reaches PR-AUC 0.265 and recovers 373 gold test edges that the rule misses. The stack wins on ranking, not at the operating cut (mention F1 0.339 versus stack F1 0.328). The reported DAG is induced from scaled B5, not from the stack: B1 was scored only on validation and test, so a full-pool stack graph is not available. Temperature scaling is implemented and almost flat, as a no-bias map must be at a 2% base rate. Directional constraints applied to the same scaled B5 scores produce a DAG of 8,906 edges. E-test pair F1

for membership in that DAG is 0.163, which is not the pair-level B5 F1 of 0.203. The DAG recovers 15.4% of unique gold edges (31.8% of gold-in-pool) at precision 0.100; its closure reaches 21.0% of unique gold.

Two properties of the data still bound every graph claim. S2 covers 48.3% of gold edges, so a perfect classifier on the current pool would still miss more than half of the ontology; adding single-token label overlap would raise that ceiling to 74.9% at a much larger pool. The 5,627 documents are ESCO descriptions of the same skills.

The 240-pair human check is a protocol pilot by the authors. In-lab κ=0.79 is not an independent-rater estimate; the four-way type field is unevaluated; and the ten-percent budget is 24 pairs. The annotators saw retrieved spans, so Table 9 may favor B3 and B5 over labels-only B1. We omit LectureBank from the result tables because the public files contain topic titles, not lecture slides, and a title-only probe is not a second-corpus test.

The tool is the induction pipeline—candidate generation, verbatim taxonomy cards, a validation threshold, C1–C5, a mention-plus-LLM stack, and a review queue—together with an explicit account of what each stage contributes and what it costs.

8. Limitations

Several measurements remain out of reach on this dump. The Wikidata P4644 map yields only 523 sitelinks and five gold positives with both endpoints linked, so a Wikipedia RefD ranking is not available; the mention-graph analog does not improve on the one-line mention rule. Scoring the 77,507 train-pool pairs lets 501 of 1,605 in-pool train gold edges enter B7, yet C3 on that denser graph also deletes test true positives (Etv recall 0.465 to 0.326). The 48.3% S2 ceiling is unchanged.

A larger human sample labeled by raters who are not the authors would be needed before t_{ij} can be scored as a four-way type: the second 240-pair sheet produced a single `transitive` judgment. Those raters should not see retrieved spans if the aim is to avoid a bias toward B3/B5. A pre-registered selective-review test is likewise still open. The stack was fit and thresholded on the same validation split, and B1 was not scored on the 77,507 train pairs, so a stack-induced DAG on the full pool is not reported.

We also lack a slice whose gold edges touch skills introduced after the model cutoff. On ESCO-PrereqSkill the v1.1.1 URI difference contains no gold positive; label permutation already drives AUC to chance. Finally, the shuffled-card placebo uses a gold-complete subsample of 5,015 pairs rather than a second pass over all 49,608 test pairs. Every B3 test span is already the card of skill A or B.

9. Conclusion

We asked when taxonomy-card formatting and directional constraints help an LLM pair scorer on a skill taxonomy. On ESCO split E, repeating the skill card as a

retrieved span helps `gpt-4o-mini` rank pairs, lowers its direction accuracy, and lowers Qwen ranking on the same test split. A mention rule has higher standalone PR-AUC than the LLM conditions; stacking that rule with B1 and B5 raises PR-AUC from 0.151 to 0.265. Calibration is nearly flat, as a no-bias temperature map must be at a 2% base rate. The reported graph—temperature-scaled B5 scores, then C2 and C3—yields a DAG with precision 0.100 and full-gold recall 0.154.

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The authors labeled the 240-pair sample (two annotators and one adjudicator) without seeing gold or model scores; this is a protocol pilot, not an independent rater study.

Data and Code Availability

ESCO-PrereqSkill is public. [5] A reproduction pack accompanying this manuscript contains the conversion and scoring scripts, the S4 prompts, the configuration file (seed 42, unseen-concept fraction 0.2), the list of held-out skill identifiers, the two 240-pair human sheets (filled forms, overlap labels, and adjudicated gold), and the pair scores ($p_{A\to B}$, $p_{B\to A}$, gold) needed to recompute the E-test tables under B1/B3/B5 and the Qwen 2.5 7B conditions. The pack is public at `https://github.com/tuanymc/p2a-prerequisite-graphs`. The full prompt-and-span logs will be posted with the camera-ready version.

Conflict of Interest

The authors declare that they have no conflict of interest.

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Appendix A. Scoring Prompts

The system message is the same for B1, B3, and B5:

```
You judge directed prerequisite relations.
Return JSON only. Do not invent quotations.
Cite only provided evidence_ids.
Fields:
- p_A_to_B, p_B_to_A: numeric floats in [0,1],
  never strings or labels
- relation_type: direct | transitive | none | ambiguous
- direction: A_to_B | B_to_A | none
- confidence_raw: numeric float in [0,1]
- rationale: short string
- evidence_ids: array of provided ids
```

The user message always begins with the two skill records (`skill_id`, preferred label, and description). B5 then inserts the taxonomy paths `Taxonomy A:` and `Taxonomy B:`. B3 and B5 append retrieved windows as `Evidence A→B:` and `Evidence B→A:`, each line `[source_id] span`. B1 replaces those blocks with the sentence "No retrieved evidence. Do not fabricate spans." Every variant ends with "Decide both directions. JSON only." Decoding temperature is 0. The model is asked for both directions in one call.

Appendix B. Token Cost and Wall-Clock Time

A 400-pair sample of B5 prompts for `gpt-4o-mini` gave a mean of 401 input tokens per pair (90th percentile 448). We budget 160 output tokens per pair from the JSON schema. At the October 2026 OpenAI list price of \$0.15 / \$0.60 per million input / output tokens, one pass over 58,219 pairs (E validation + E test) has an expected list-price cost of about \$9 (range \$7–\$12 in the same estimate); the additional 77,507 E-train pairs are about \$12 on the same budget (together 135,726 scored B5 pairs). Taxonomy lines in B5 add only about two lines relative to B3. The parallel score loop took about 3.8 h (B1), 5 h (B3), and 4.8 h (B5 val+test) of wall-clock time, plus about 6.9 h for E-train B5. Eight B5 requests that had initially been recorded as p=0 were retried on the same model. Per-request latency was not logged; the hours above are the end-to-end loop, not a single-thread latency. Qwen 2.5 7B was scored on E validation (8,611 pairs) and E test (49,608 pairs) through an OpenAI-compatible API; that token bill was not kept. Failed test requests (four on B3, 57 on B5) were retried. The mention-plus-LLM stack needs both B1 and B5 scores, so its token cost is about twice that of a single prompt variant on the same pairs.

References

1. K. Xiao, L. Zheng, X. He, M. Zhang, Z. Huang and Y. Zhang, LECDPR: LLM enhancement and concept-document interactive modeling for prerequisite relation prediction, in *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence IJCAI-2026*, (International Joint Conferences on Artificial Intelligence Organization, September 2026), pp. 5092–5100.

2. X. Cheng, J. Wang, C. He, R. Dong and Q. Guan, ProPRL: Property-aware prerequisite relation learning in educational knowledge graphs arXiv:2608.03006, (2026).
3. X. Li, F. Min, M. Li, C. Zhang and Z. Wang, AutoPRE: Discovering concept prerequisites with LLM agents, in *Natural Language Processing and Chinese Computing* (Springer Nature Singapore, November 2025), pp. 199–211.
4. X. Cheng, Z. Zhang, J. Wang, L. Fang, C. He, Q. Guan, S. Pan and W. Luo, GraphRAG-induced dual knowledge structure graphs for personalized learning path recommendation, *Proceedings of the AAAI Conference on Artificial Intelligence* **40** (March 2026) 14610–14620.
5. N. L. Le and M.-H. Abel, How well do LLMs predict prerequisite skills? zero-shot comparison to expert-defined concepts, in *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (IEEE, October 2025), pp. 7245–7250.
6. Y. Zhu and H. Zamani, Predicting prerequisite relations for unseen concepts, in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2022), pp. 8542–8548.
7. V. Ramachandran, M. Vedanta and S. Khetarpaul, Cortex: A retrieval-augmented framework for career-aligned learning paths, *Machine Learning* **115** (September 2026).
8. C. Liang, Z. Wu, W. Huang and C. L. Giles, Measuring prerequisite relations among concepts, in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2015), pp. 1668–1674.
9. C. Liang, J. Ye, Z. Wu, B. Pursel and C. Giles, Recovering concept prerequisite relations from university course dependencies, *Proceedings of the AAAI Conference on Artificial Intelligence* **31** (February 2017).
10. Y. Yang, H. Liu, J. Carbonell and W. Ma, Concept graph learning from educational data, in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining WSDM 2015*, (ACM, February 2015), pp. 159–168.
11. H. Liu, W. Ma, Y. Yang and J. Carbonell, Learning concept graphs from online educational data, *Journal of Artificial Intelligence Research* **55** (April 2016) 1059–1090.
12. L. Pan, C. Li, J. Li and J. Tang, Prerequisite relation learning for concepts in MOOCs, in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Association for Computational Linguistics, 2017), pp. 1447–1456.
13. S. Roy, M. Madhyastha, S. Lawrence and V. Rajan, Inferring concept prerequisite relations from online educational resources, *Proceedings of the AAAI Conference on Artificial Intelligence* **33** (July 2019) 9589–9594.
14. W. Lu, Y. Zhou, J. Yu and C. Jia, Concept extraction and prerequisite relation learning from educational data, *Proceedings of the AAAI Conference on Artificial Intelligence* **33** (July 2019) 9678–9685.
15. K. Xiao, Y. Bai and Z. Wang, Extracting prerequisite relations among concepts from the course descriptions, *International Journal of Software Engineering and Knowledge Engineering* **32** (April 2022) 503–523.
16. J. Sun, Y. He, Y. Xu, J. Sun and G. Sun, A learning-path based supervised method for concept prerequisite relations extraction in educational data, in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management CIKM '24*, (ACM, October 2024), pp. 2168–2177.
17. Y. Zhang, X. Qu, S. Liu, Y. Pang and X. Shang, Multiscale Weisfeiler-Leman directed graph neural networks for prerequisite-link prediction, *IEEE Transactions on Knowledge and Data Engineering* **37** (June 2025) 3556–3569.

18. D. Mazumder, J. H. Paik and A. Basu, A graph neural network model for concept prerequisite relation extraction, in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management CIKM '23*, (ACM, October 2023), pp. 1787–1796.
19. J. Xia, M. Li, Y. Tang and S. Yang, Course map learning with graph convolutional network based on AuCM, *World Wide Web* **26** (July 2023) 3483–3502.
20. I. Li, A. R. Fabbri, R. R. Tung and D. R. Radev, What should I learn first: Introducing LectureBank for NLP education and prerequisite chain learning, *Proceedings of the AAAI Conference on Artificial Intelligence* **33** (July 2019) 6674–6681.
21. P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel and D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks arXiv:2005.11401, (2020), Neural Information Processing Systems.
22. P. Chen, Y. Lu, V. W. Zheng, X. Chen and B. Yang, KnowEdu: A system to construct knowledge graph for education, *IEEE Access* **6** (2018) 31553–31563.
23. F. Chiarello, G. Fantoni, T. Hogarth, V. Giordano, L. Baltina and I. Spada, Towards ESCO 4.0 – is the European classification of skills in line with Industry 4.0? a text mining approach, *Technological Forecasting and Social Change* **173** (December 2021) p. 121177.
24. M. Hosseinioun, F. Neffke, L. Zhang and H. Youn, Skill dependencies uncover nested human capital, *Nature Human Behaviour* **9** (February 2025) 673–687.
25. Y. Dai, M. Yoshikawa and K. Sugiyama, Prerequisite-aware course ordering towards getting relevant job opportunities, *Expert Systems with Applications* **183** (November 2021) p. 115233.
26. N. L. Le, M.-H. Abel and B. Laforge, From learning resources to competencies: LLM-based tagging with evidence and graph constraints, in *Artificial Intelligence in Education* (Springer Nature Switzerland, June 2026), pp. 616–631.
27. C. Alzetta, I. Torre and F. Koceva, Annotation protocol for textbook enrichment with prerequisite knowledge graph, *Technology, Knowledge and Learning* **29** (March 2024) 197–228, Published online 21 September 2023.
28. C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, On calibration of modern neural networks arXiv:1706.04599, (2017), International Conference on Machine Learning.